

3D Surface Reconstruction Using Point Clouds from RGB-D Image Data

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Outline



Introduction



Motivation



Objectives



Methodology



Dataset

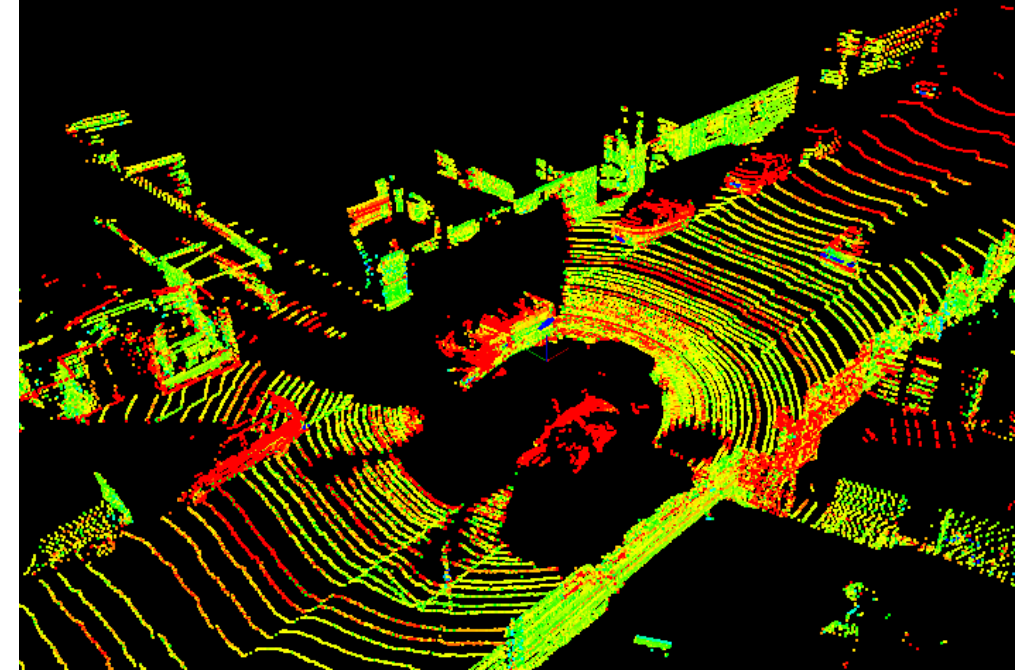
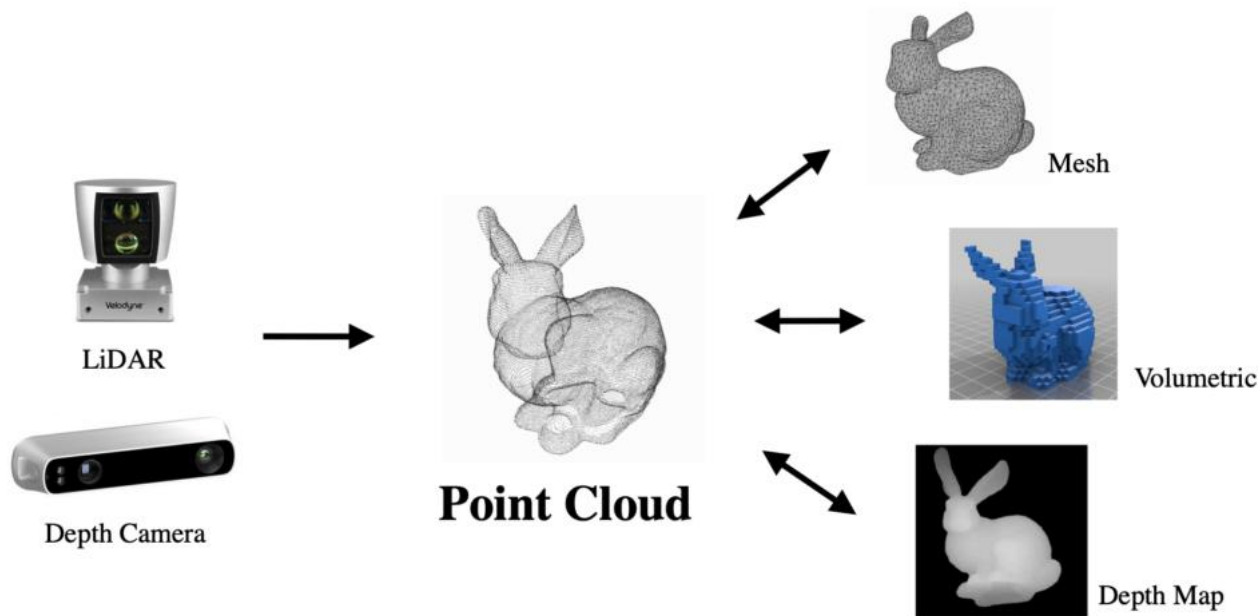


Results and Evaluation



Introduction

- What is a Point Cloud?



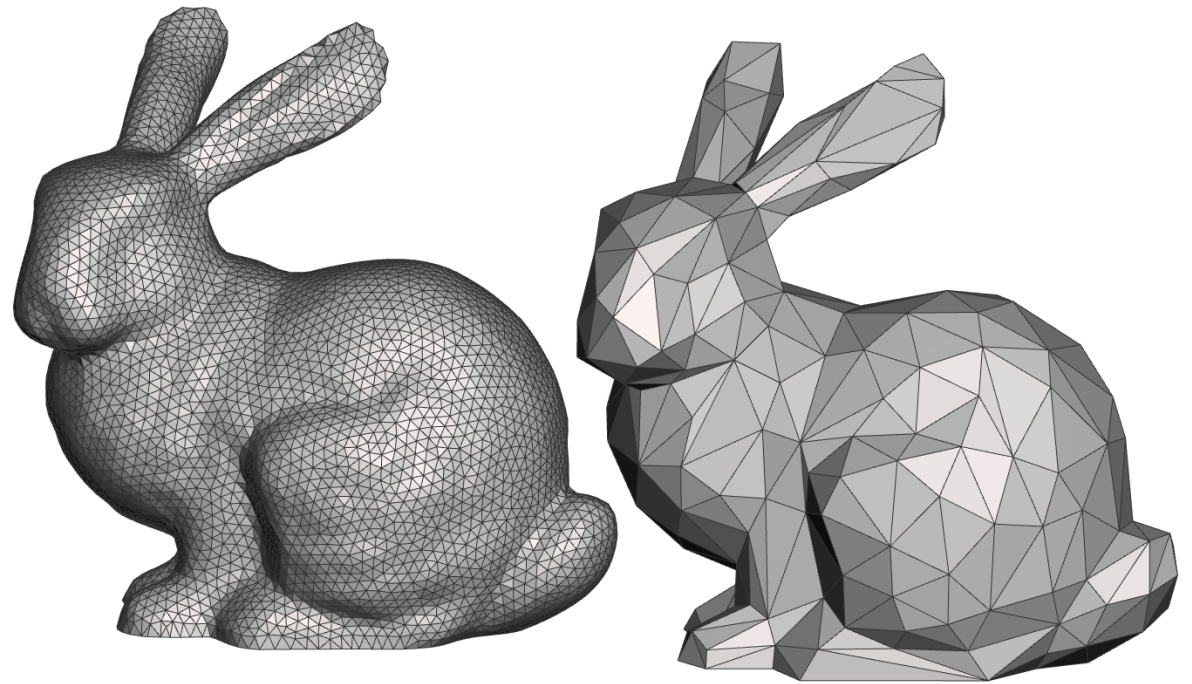
- A point cloud is a set of data points in a 3D coordinate system. Each point represent a precise location in space
- Point clouds create a three-dimensional representation of an object or area.
- They are generated by 3D scanners that use lasers, cameras or sensors.
- Each point in a point cloud has x,y,z coordinates that create a 3D model of the object in the real environment³

Introduction

- How to create a **digital copy** of a **real object**?



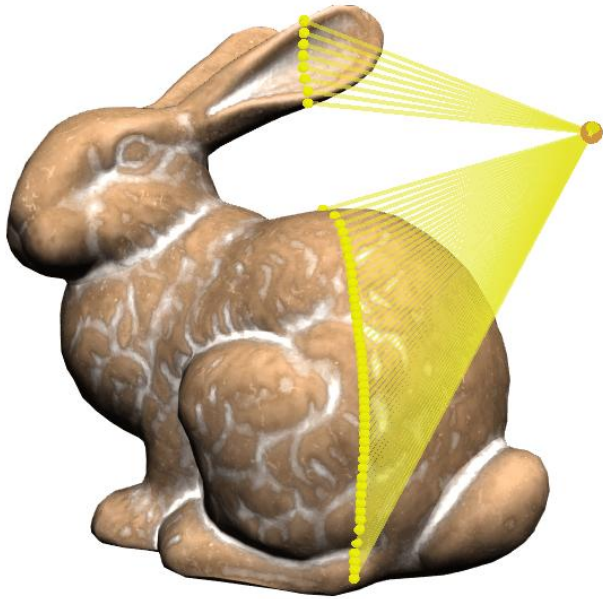
Real Object



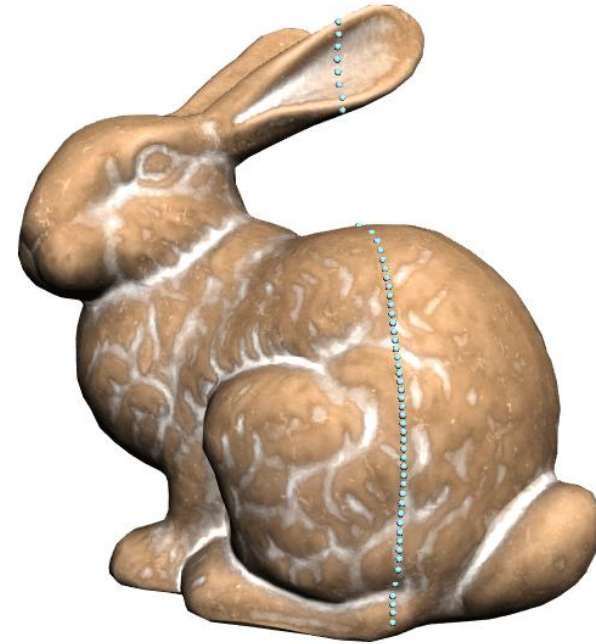
Digital Copy

Introduction

- Scan object to create a **3D point cloud**



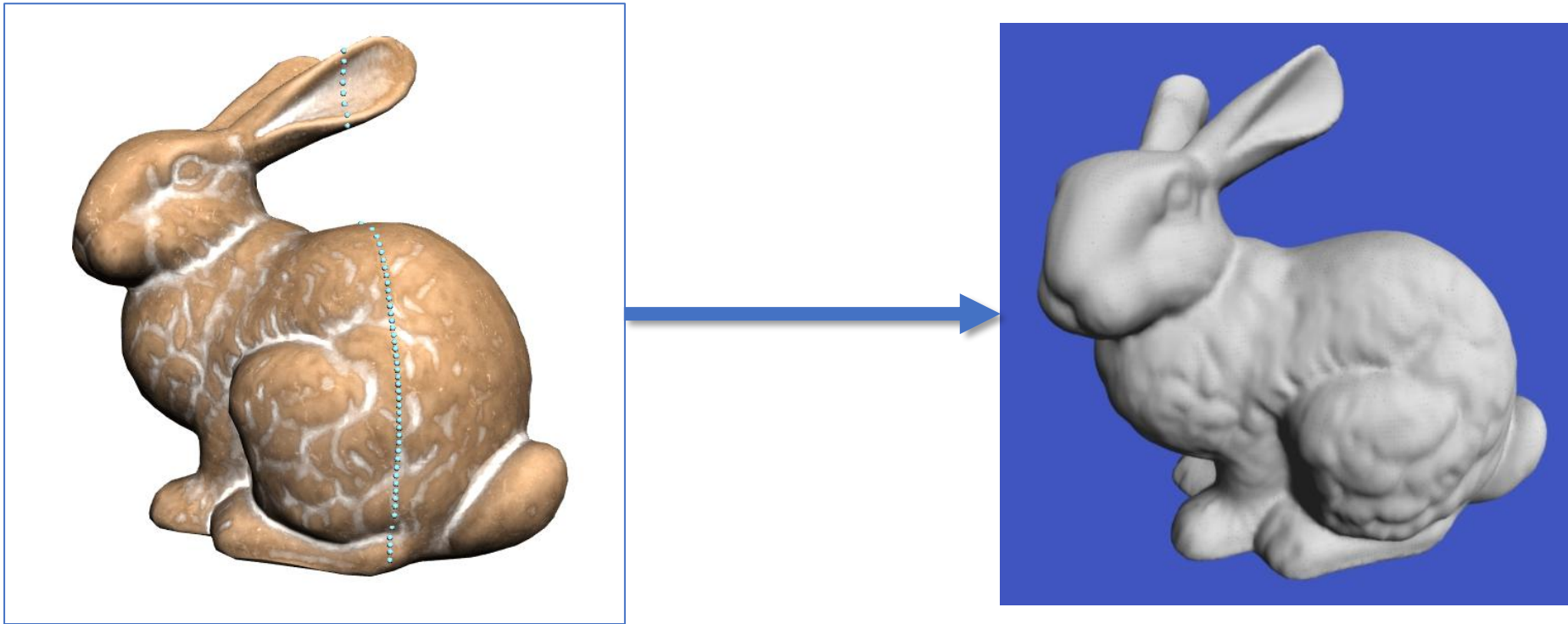
Scanning the Object



3D Point Cloud

Introduction

- Approximate the **surface** given the point cloud

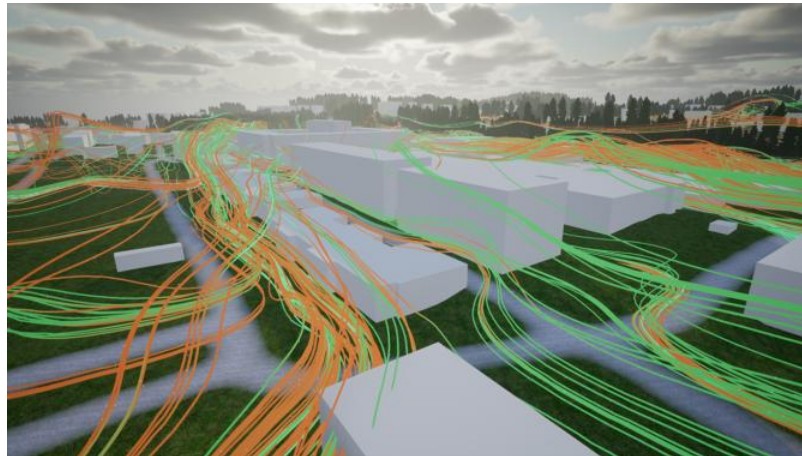


Motivation

- Visualization
- Analysis
 - Mechanical Engineering
 - Environmental Science
 - Urban Planning
 - Robotics
 - Autonomous Driving
 - AR/VR
 - Gaming
 - Architecture



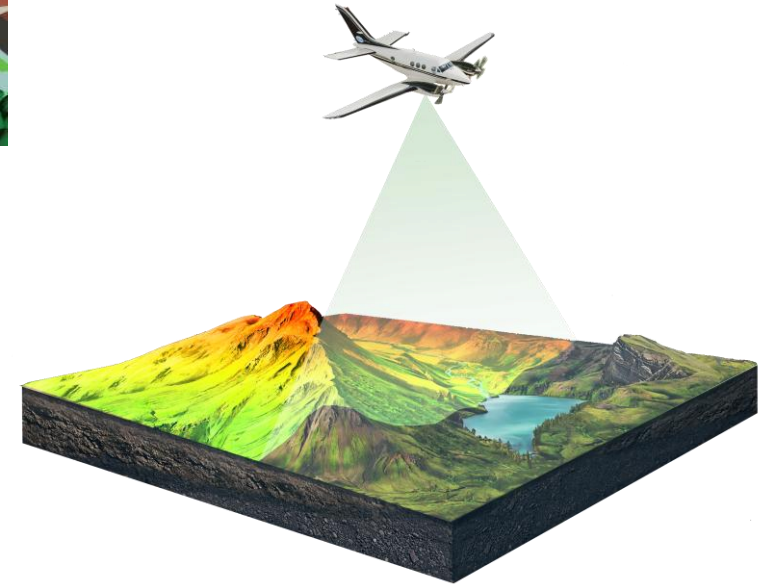
Surface Reconstruction



Wind Simulation



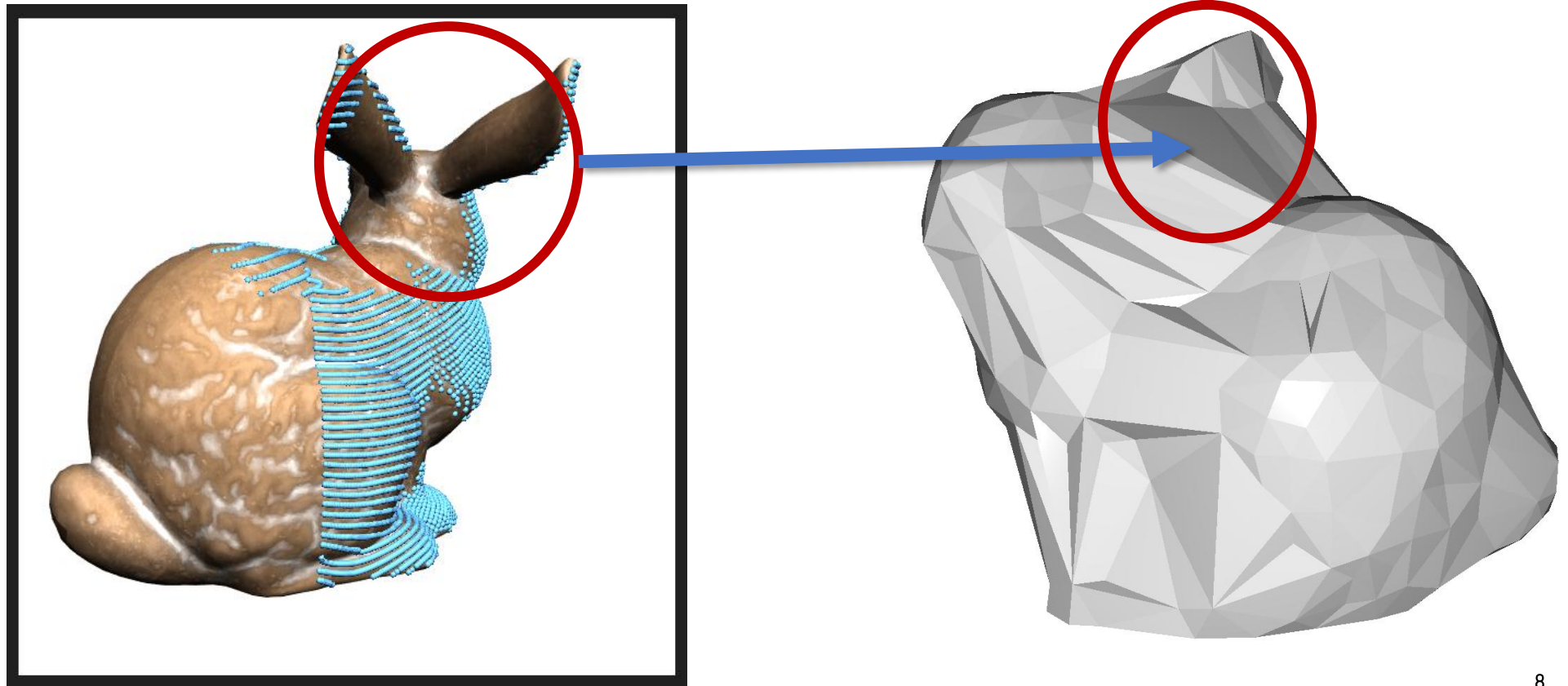
Point Clouds Visualization in VR



Point Cloud Acquisition

Objective

Problem: We do not know what lies between the points!



Objective

Can we learn to reconstruct surfaces from these point clouds?



Objective

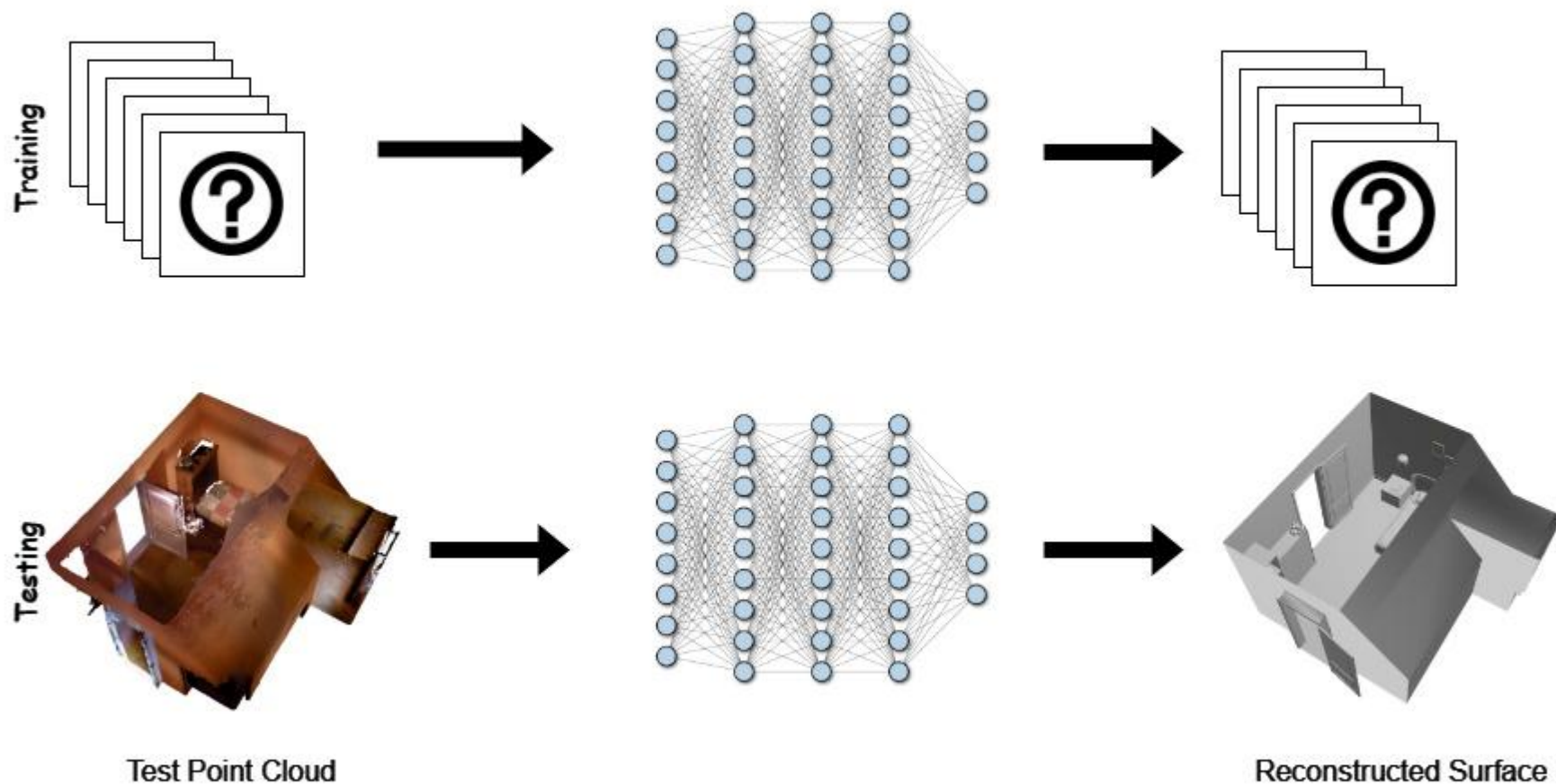
Can we learn to reconstruct surfaces from these point clouds?

Difficulties:

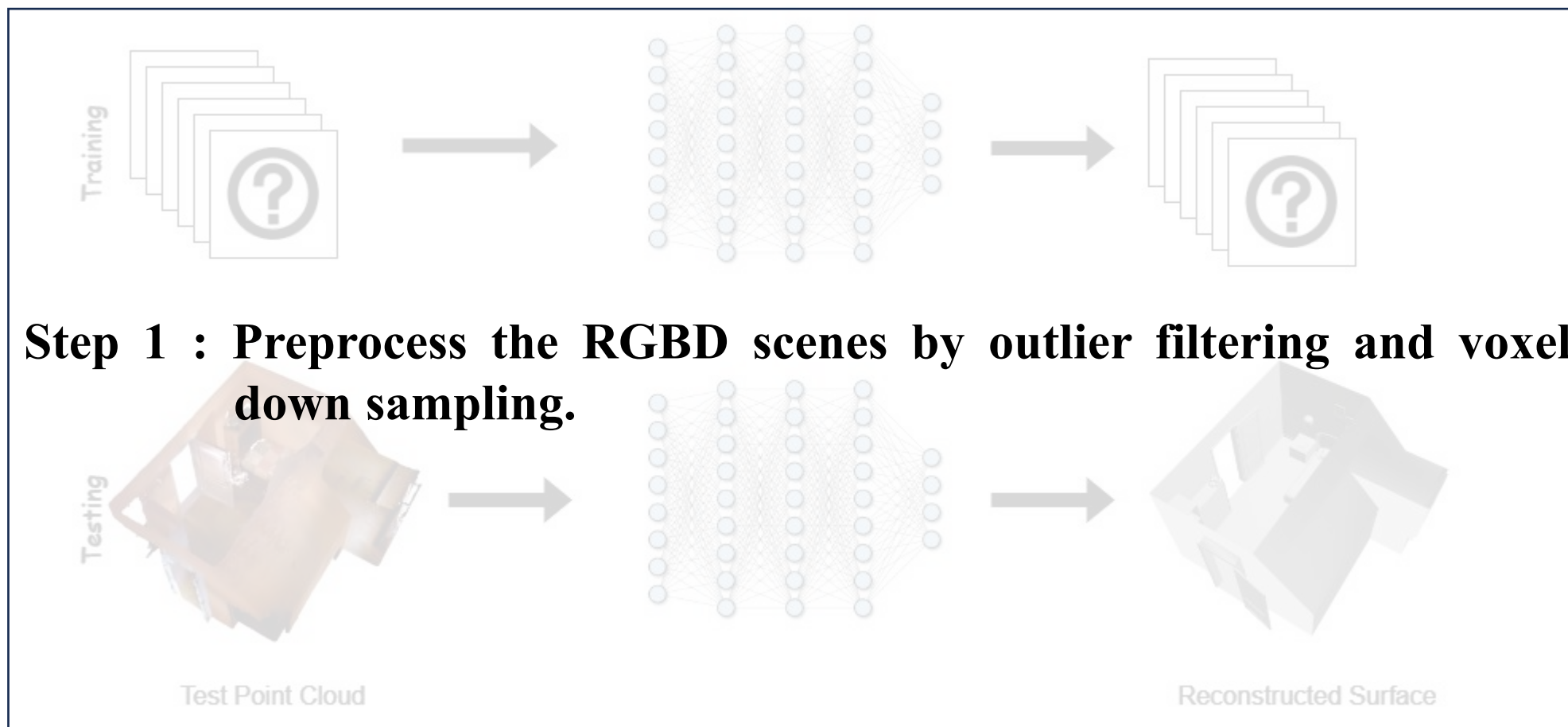
- Point cloud defects
 - Missing data
 - Outliers
 - Noise
 - Non-uniform density
- Multiple modalities
 - LiDAR
 - Multi-View Stereo (MVS)
- Large scale ($>1\text{M}$ points)

Approach

Formulate surface reconstruction as a supervised learning problem

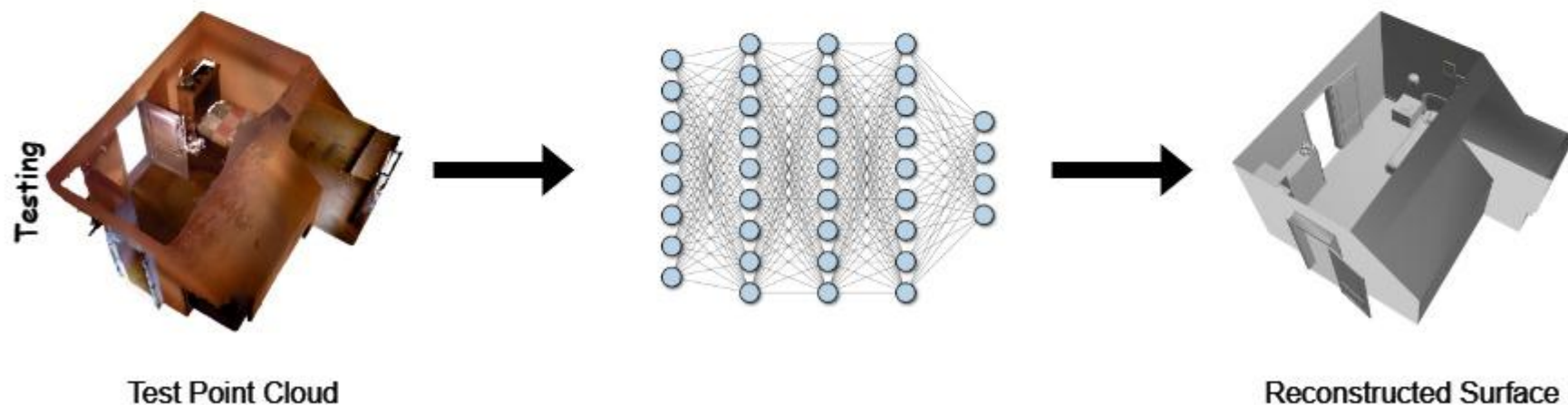


Approach



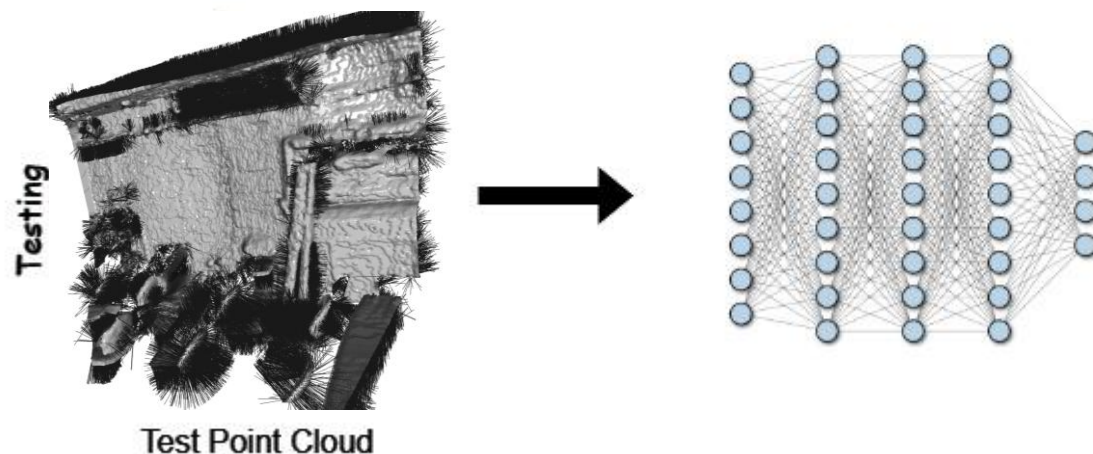
Approach

Step 2: Using existing architectures for learning-based surface reconstruction



Approach

Step 3: Incorporate visibility information into learning-based surface reconstruction



Dataset

SUN RGB-D: A RGB-D Scene Understanding Benchmark Suite

The SUN RGBD dataset is captured by four different sensors and contains 10335 real RGB-D images of room scenes.

- Each RGB image has a corresponding depth and segmentation map.
- The training and testing sets contain 5285 and 5050 images, respectively.
- The whole dataset is densely annotated and includes 146,617 2D polygons.

Images were taken using 4 different sensors

	Real Sense	Xtion	Kinect v1	Kinect v2
weight (pound)	0.077	0.5	4	4.5
size (inch)	5.2×0.25×0.75	7.1×1.4×2	11×2.3×2.7	9.8×2.7×2.7
power	2.5W USB	2.5W USB	12.96W	115W
depth resolution	628×468	628×468	628×468	512×424
color resolution	1920×1080	640×480	640×480	1920×1080

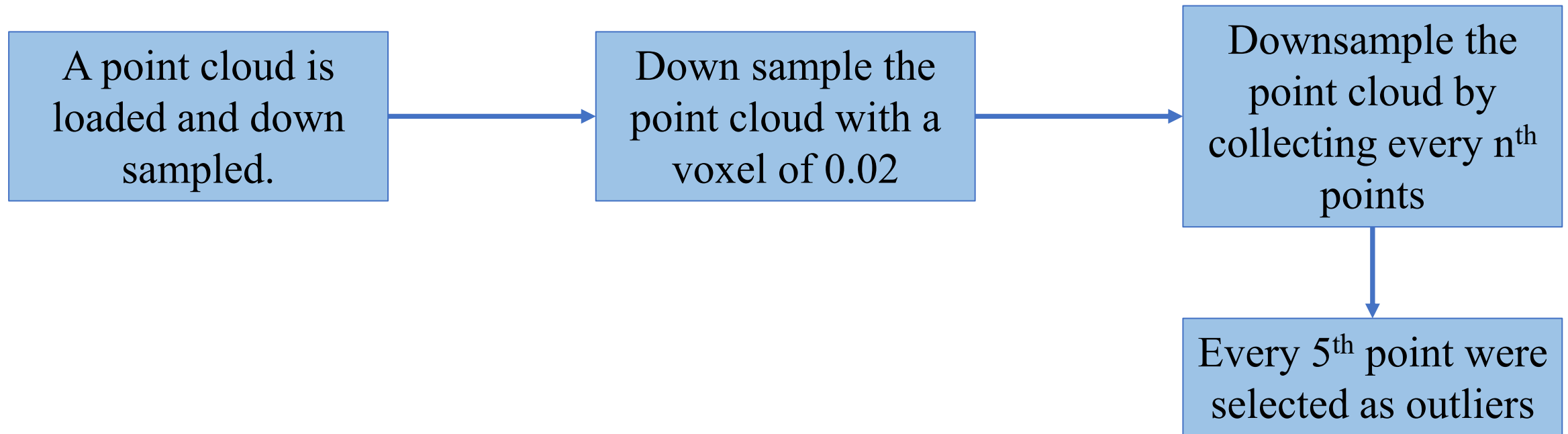
Dataset Usage

- Only 500 images were used in training and 100 in testing.
- I have used the RGB color image, and the depth image combined to get the point cloud as the input.
- The dataset includes watertight ground truth surfaces for each object.
- Xtion and Kinect v1 sensor data was only used as they had same color and depth image resolutions.
- ScanNet: Richly-annotated 3D Reconstructions of Indoor Scenes dataset was also used in model training.



Step 1: Preprocess the RGBD scenes by outlier filtering and voxel down sampling.

When collecting data from scanning devices, the resulting point cloud tends to contain noise and artifacts that one would like to remove.

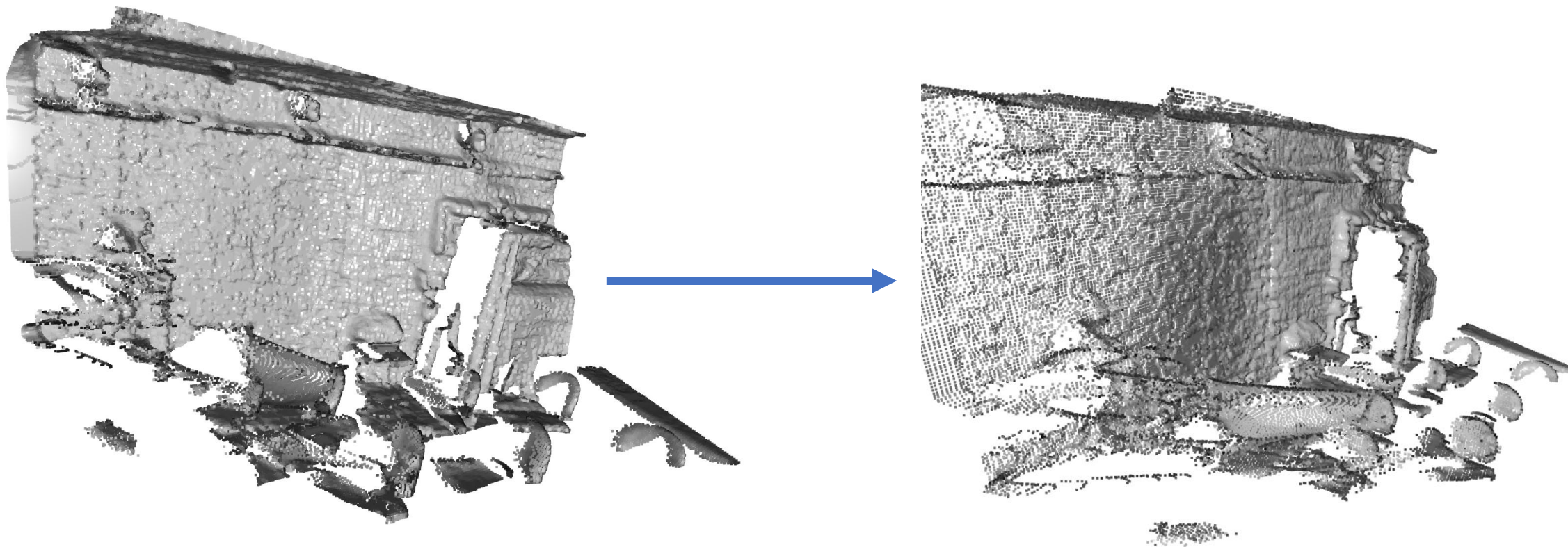


Reference:

https://www.open3d.org/docs/release/tutorial/geometry/pointcloud_outlier_removal.html

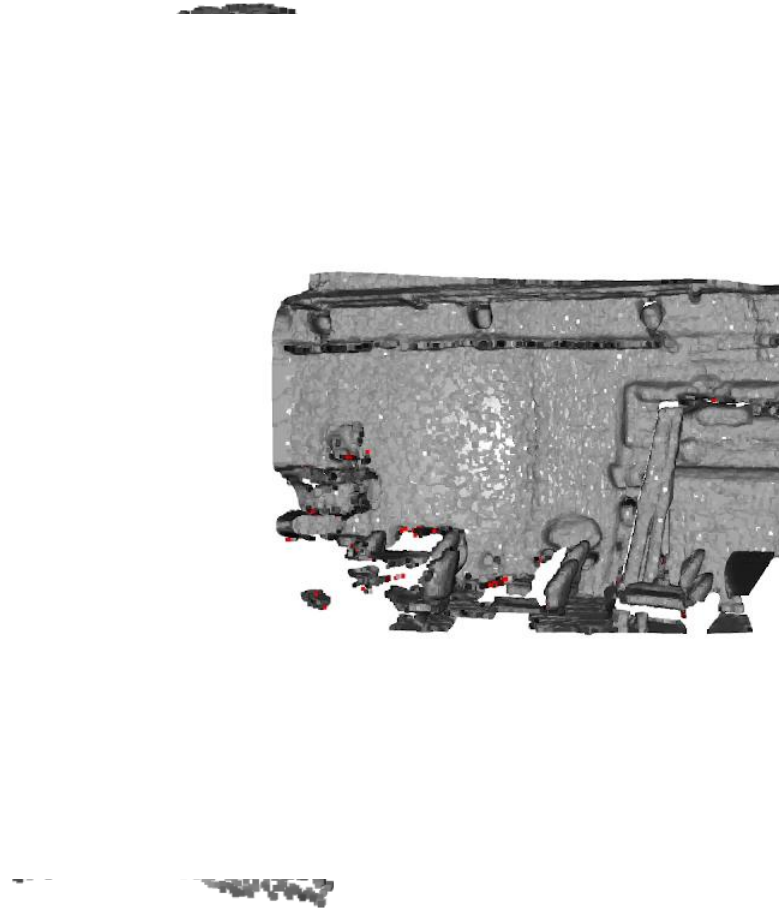
A point cloud is loaded
and down sampled.

Down sample the point
cloud with a voxel of 0.02



Downsample the point cloud
by collecting every n^{th} points.

Every 5th point were
selected as outliers



Step 2: Using Existing Architectures for Learning-Based Surface Reconstruction

Traditional methods

- Screened Poisson Surface Reconstruction (SPSR) [Kazhdan et al. 2013]
- Robust and Efficient Surface Reconstruction (Labatut et al.) [Labatut et al. 2009]

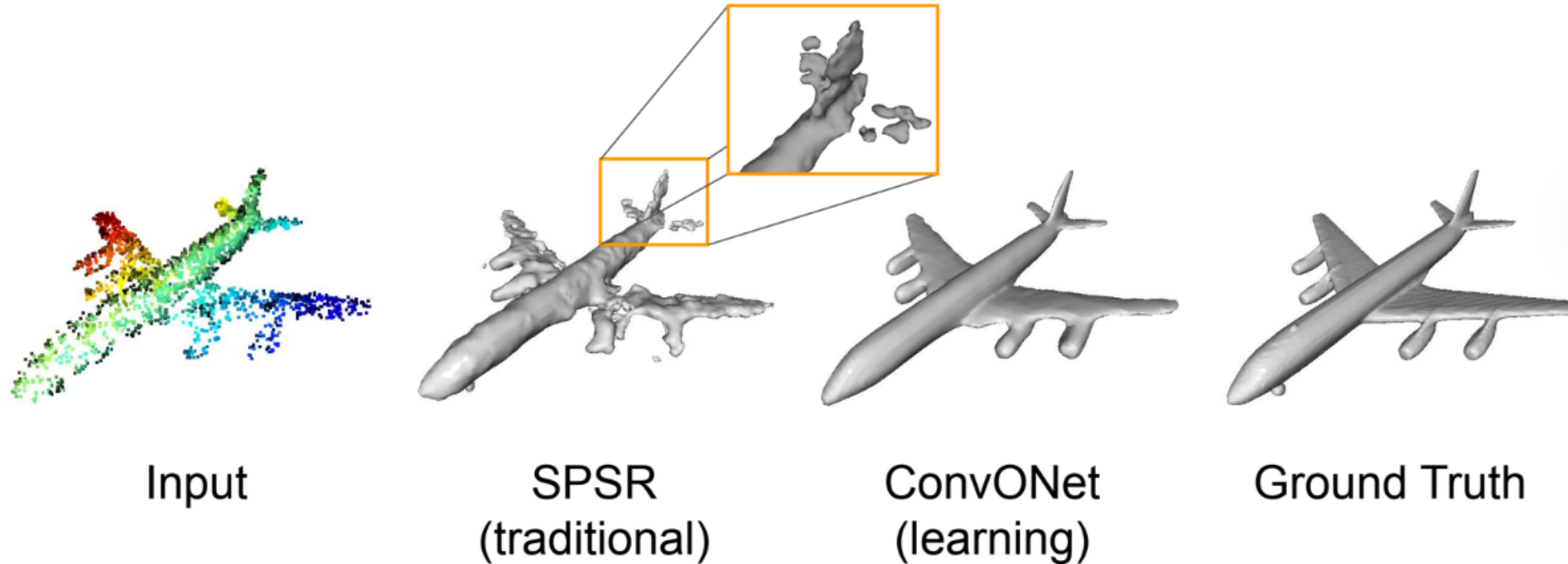
Learning-based methods

- Convolutional Occupancy Networks (ConvONet) [Peng et al. 2020]
- Shape-As-Points (SAP) [Peng et al. 2021]
- Point Convolution for Surface Reconstruction (POCO) [Boulch & Marlet 2022].

Reference: <https://arxiv.org/pdf/2301.13656.pdf>

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds

Known sampling defects and known shape categories

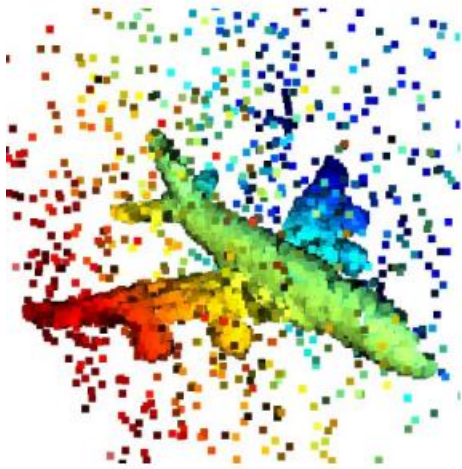


Learning-based methods provide more accurate reconstructions

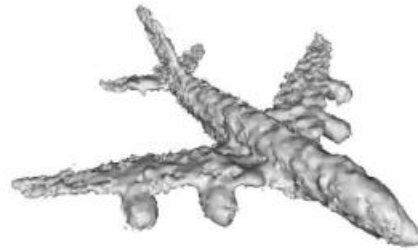
Reference: Deep Surface Reconstruction from Point Clouds with Visibility Information

Paper: <https://arxiv.org/abs/2202.01810>

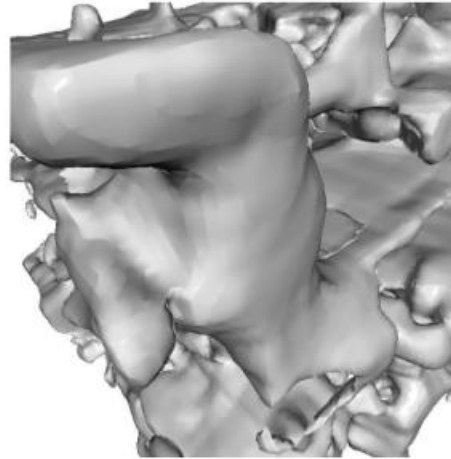
Unknown sampling defects and known shape categories



Input



SPSR
(traditional)



ConvONet
(learning)



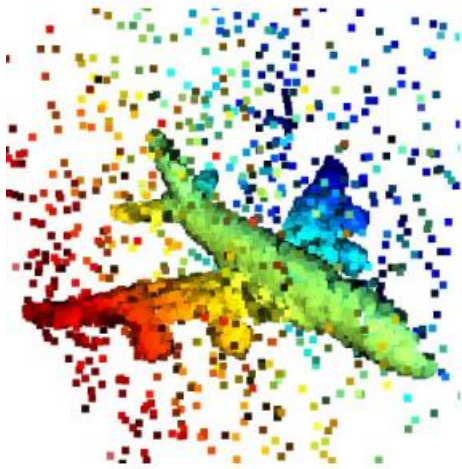
Ground Truth

Traditional methods provide more accurate reconstructions

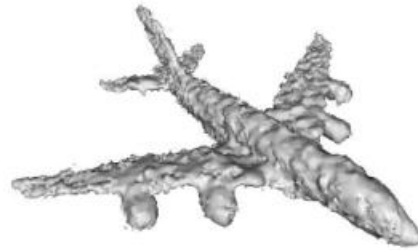
Reference: Deep Surface Reconstruction from Point Clouds with Visibility Information

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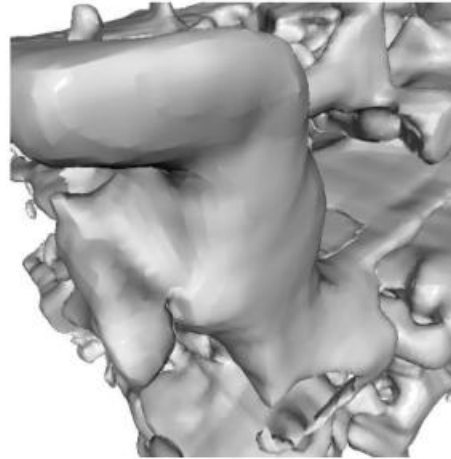
Known sampling defects and unknown shape categories



Input



SPSR
(traditional)



ConvONet
(learning)



Ground Truth

Traditional methods provide more accurate reconstructions

Reference: Deep Surface Reconstruction from Point Clouds with Visibility Information

Paper: <https://arxiv.org/abs/2202.01810>

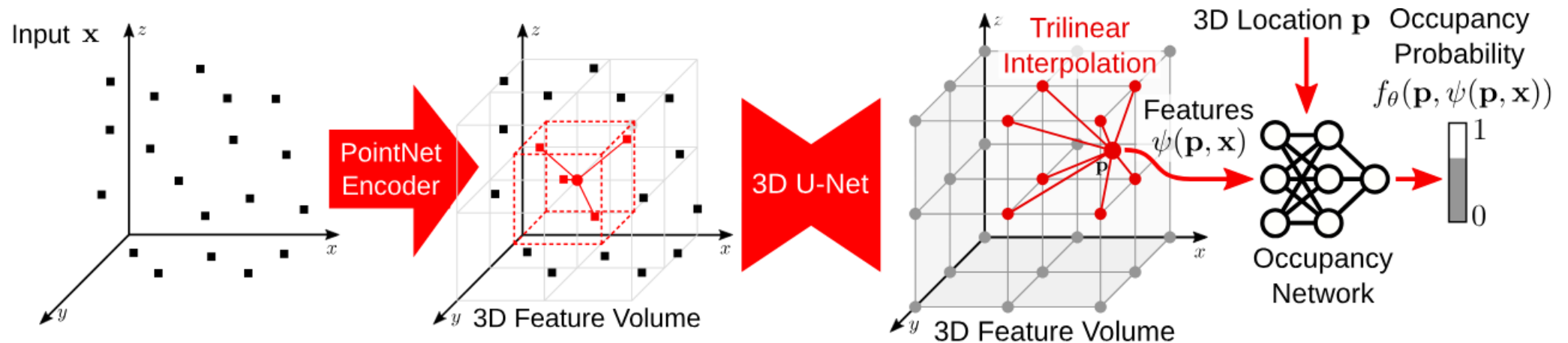
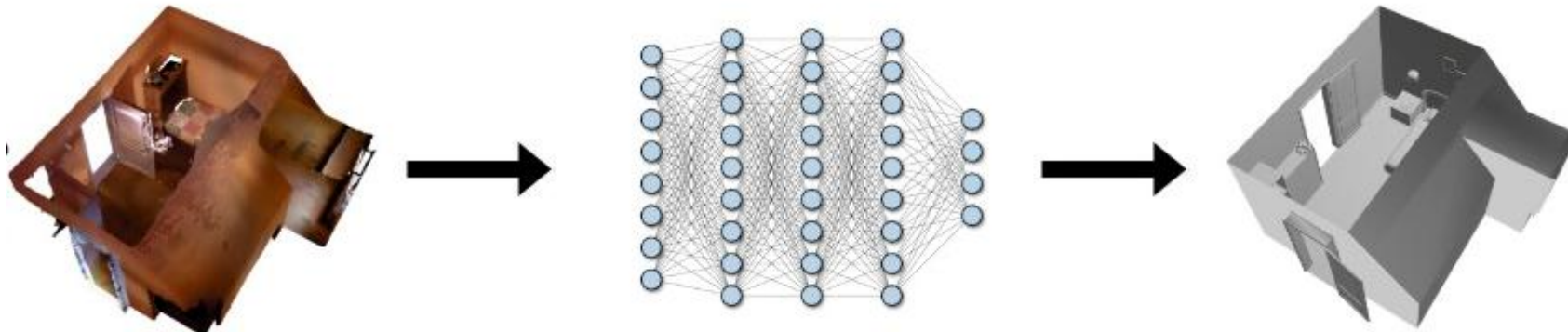
Conclusions from the Paper

Learning-based surface reconstruction...

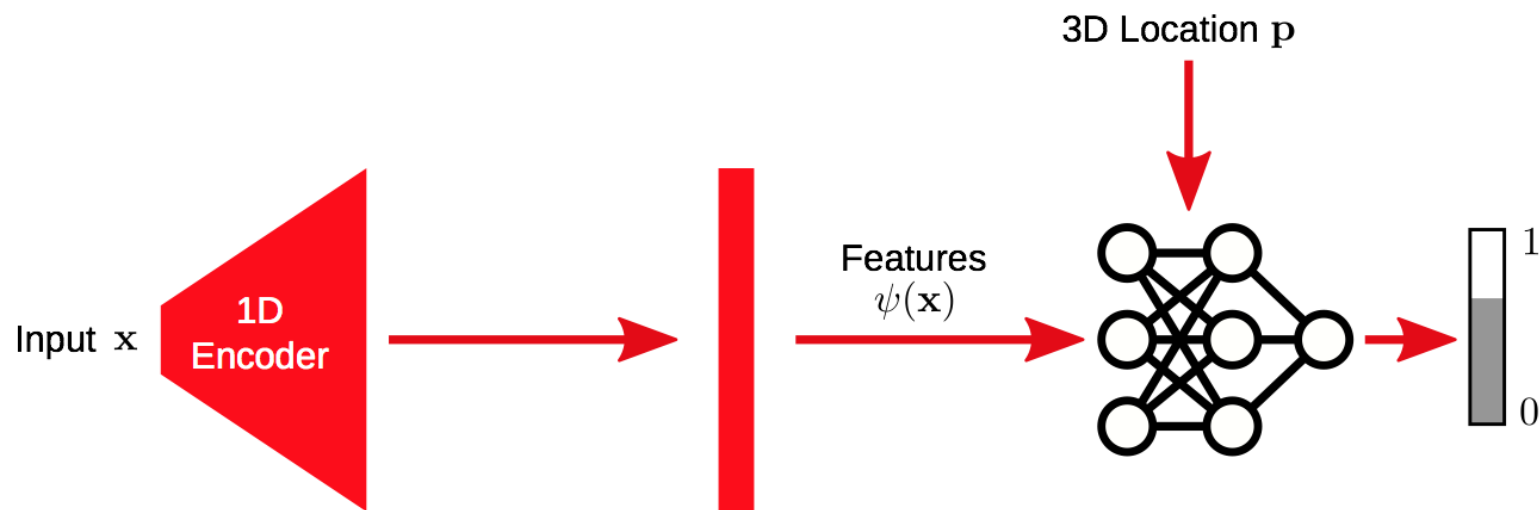
- outperforms traditional reconstruction methods when sampling and shape categories are known
- does not work well for unknown sampling defects.
needs a large training set with complex shapes

Reference: Deep Surface Reconstruction from Point Clouds with Visibility Information
Paper: <https://arxiv.org/abs/2202.01810>

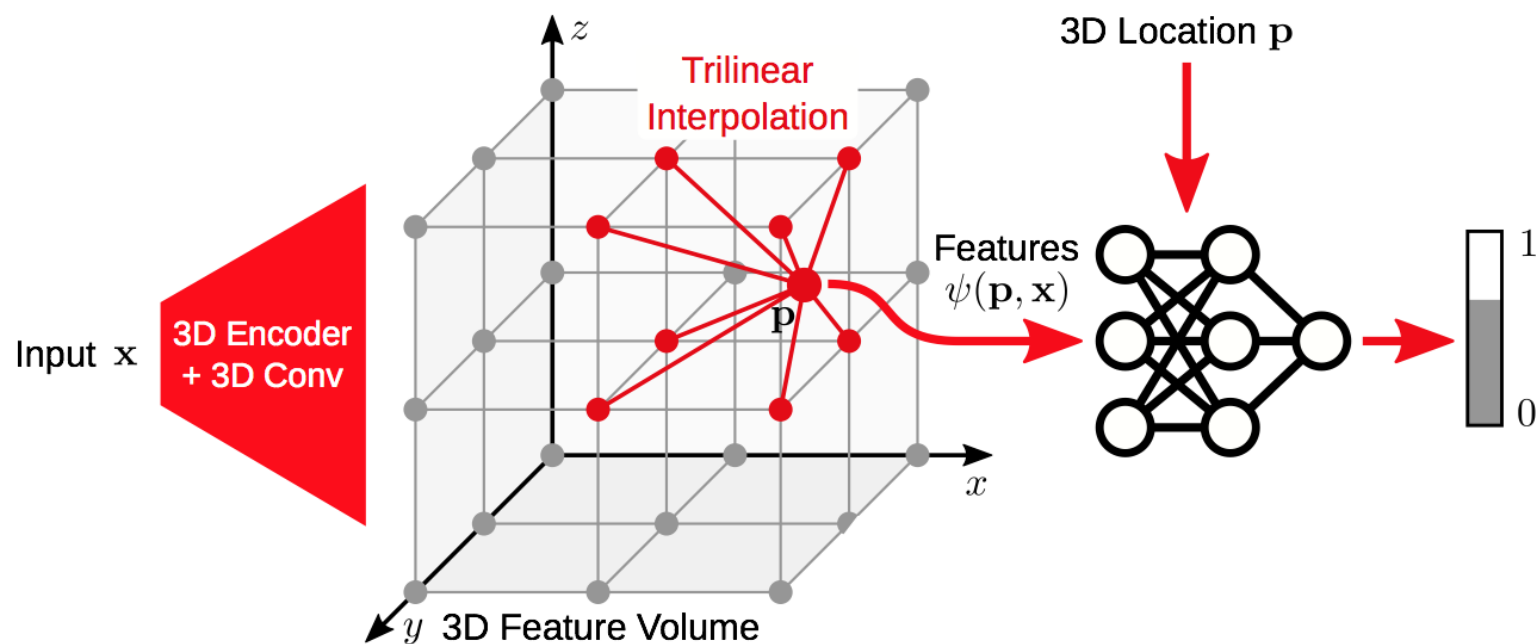
Convolutional Occupancy Networks (ConvONet)



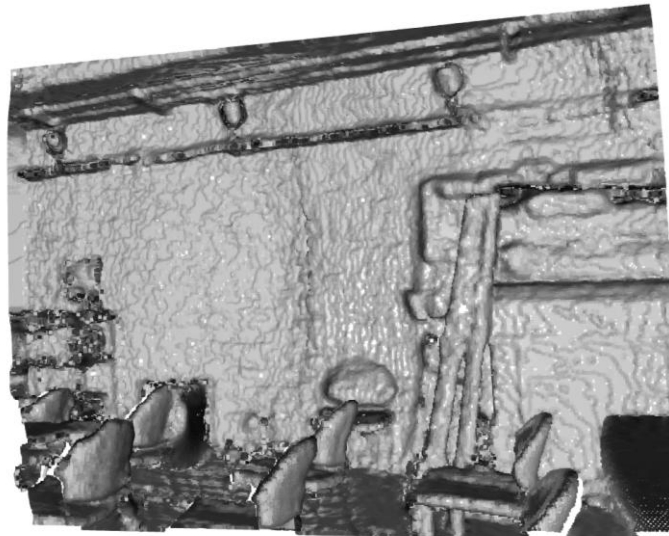
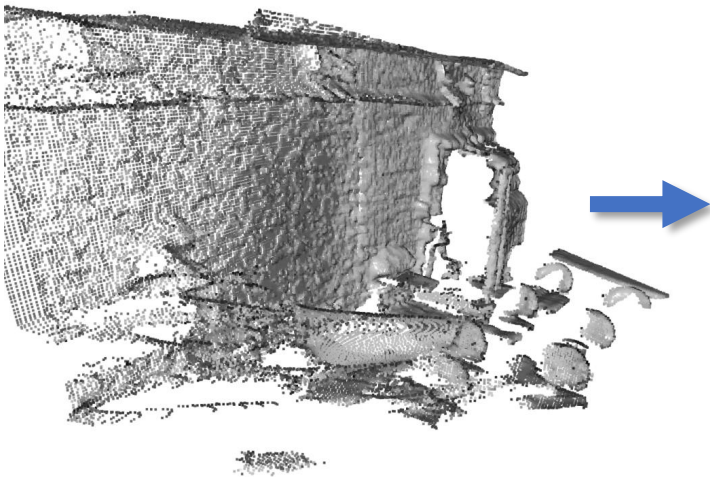
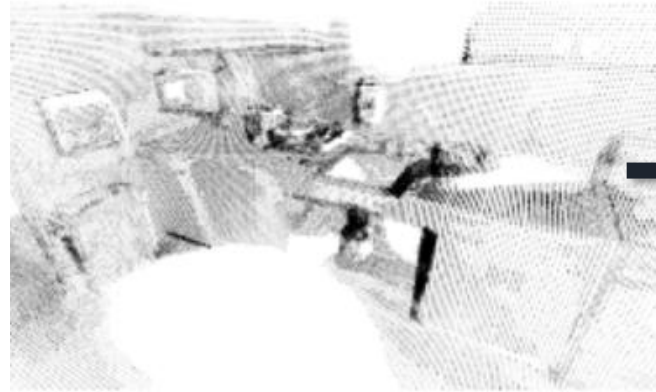
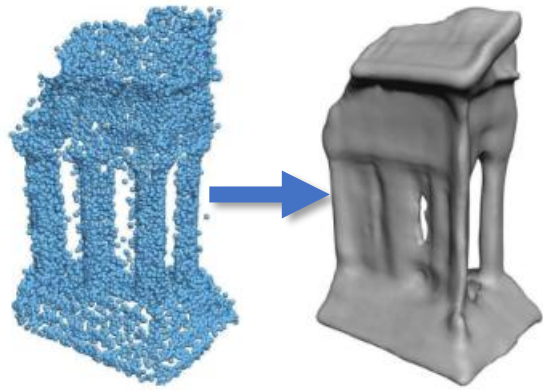
Occupancy Networks



Convolutional Occupancy Networks



Results



Evaluation

Geometric metrics.

We evaluate reconstructions with the

- volumetric Intersection over union (IoU)
- symmetric Chamfer distance (CD)
- normal consistency (NC)

Normal Consistency (NC)

A measure of how consistent the orientations of surface normals are across neighboring points or regions of a 3D model.

$n(x)$ is the unit normal of a point x on a mesh, and $\langle u, v \rangle$ the Euclidean scalar product in $\mathbb{R}^3$.

$$\text{NC}(\mathbf{S}^g, \mathbf{S}^r) = \frac{1}{2|P^g|} \sum_{x \in P^g} \langle n(x), n(\argmin_{y \in P^r} \|x - y\|_2) \rangle + \frac{1}{2|P^r|} \sum_{x \in P^r} \langle n(\argmin_{y \in P^g} \|y - x\|_2), n(x) \rangle$$

Findings

Training

Method	Volumetric IoU (%)	Normal consistency (%)	Chamfer Distance (x10 ³) <i>Loss Function</i>
	Preprocessed Point Clouds	Preprocessed Point Clouds	Preprocessed Point Clouds
Convolutional Occupancy Network (3D)	62.38	76.47	1.666

Testing

Method	Volumetric IoU (%)	Normal consistency (%)	Chamfer Distance (x10 ³) <i>Loss Function</i>
	Preprocessed Point Clouds	Preprocessed Point Clouds	Preprocessed Point Clouds
Convolutional Occupancy Network (3D)	60.30	70.56	1.895

Challenges

- Time constraints in model training on the dataset for different models.
- Limited GPU power as it takes time in processing the images with low end GPU.

Future Work

- Experiment on more state-of-the-art learning-based networks and compare results to find more suitable network for reconstruction and compare with Augmented Surfaces.
- Incorporate visibility information into learning-based surface reconstruction and follow the same process in experimenting.